

ERPChat Complete Database Schema Dictionary

Purpose & Embedding Optimization Notice: This document serves as a semantic data dictionary of the ERPChat system schema. It is structured explicitly to optimize chunking and document retrieval via embedding models (RAG). Each table section contains fully self-contained context, including descriptions, column metadata, primary/foreign keys, and data relationships to prevent loss of context during chunk splitting.

Table 1: products

Description: Central catalog storing all core item metadata, pricing configurations, and taxation states for the ERP system.

Column Name	Data Type	Key Type	Description / Constraints
product_id	int4	Primary Key	Unique auto-incrementing identifier for each product.
sku	varchar	Unique	Stock Keeping Unit; unique alphanumeric vendor/internal code.
product_name	varchar	-	The descriptive commercial name of the product.
category	varchar	-	Classification grouping for inventory structure and reporting.
cost_price	numeric	-	Internal acquisition or manufacturing cost per unit.
unit_price	numeric	-	Standard outgoing retail or wholesale selling price.
tax_rate	numeric	-	Applicable tax rate multiplier or percentage for billing.
is_discontinued	bool	-	Flag to signify whether the product is no longer active.

Relationships & Foreign Key References:

- Referenced by `inventory.product_id` (One-to-Many)
- Referenced by `order_items.product_id` (One-to-Many)
- Referenced by `stock_movements.product_id` (One-to-Many)

Table 2: inventory

Description: Tracks real-time physical stock levels, thresholds, and allocations across various warehouse locations.

Column Name	Data Type	Key Type	Description / Constraints
inventory_id	int4	Primary Key	Unique serial identifier for the inventory record.
product_id	int4	Foreign Key	References products.product_id to link stock metrics to an active product item.
warehouse_name	varchar	-	Name or code of the specific physical warehouse location.
stock_on_hand	int4	-	Total physical items currently present in the facility.
reorder_level	int4	-	Minimum safety threshold triggering replenishment workflows.
allocated_stock	int4	-	Items reserved for confirmed orders but not yet fulfilled.

Relationships & Foreign Key References:

- Many-to-One relationship to **products.product_id**

Table 3: stock_movements

Description: Comprehensive audit trail documenting every historical change in product stock levels (receipts, shipments, transfers, adjustments).

Column Name	Data Type	Key Type	Description / Constraints
movement_id	int4	Primary Key	Unique sequential identifier for the stock event log.
product_id	int4	Foreign Key	References products.product_id specifying which item moved.
warehouse_name	varchar	-	The specific warehouse location where the change took place.
movement_type	varchar	-	Type of ledger entry (e.g., 'INBOUND', 'OUTBOUND', 'ADJUSTMENT').
quantity	int4	-	Delta size of the transaction (positive or negative count).
reference_id	int4	-	Optional link to an source order or internal ticket.
movement_timestamp	timestamptz	-	Exact standardized timestamp of database commit.

Relationships & Foreign Key References:

- Many-to-One relationship to **products.product_id**

Table 4: customers

Description: Master registry containing commercial corporate buyers, financial boundaries, geographic location routing, and active flags.

Column Name	Data Type	Key Type	Description / Constraints
<code>customer_id</code>	int4	Primary Key	Unique serial identifier for the customer portfolio.
<code>company_name</code>	varchar	-	The registered corporate business or individual client entity name.
<code>contact_name</code>	varchar	-	Primary point-of-contact handling transactions or communication.
<code>email</code>	varchar	Unique	Primary electronic mail address for account invoicing.
<code>phone</code>	varchar	-	Contact phone number records for logistics or support.
<code>credit_limit</code>	numeric	-	Maximum authorized balance allowed for purchase orders.
<code>outstanding_balance</code>	numeric	-	Current unpaid balance accrued by the account.
<code>city</code>	varchar	-	Mailing/Billing address locality detail.
<code>state</code>	varchar	-	Mailing/Billing address administrative state or province.
<code>country</code>	varchar	-	Mailing/Billing country destination marker.
<code>is_active</code>	bool	-	Status flag determining operational validity.
<code>created_at</code>	timestampz	-	Timestamp indicating the creation of the account profile.

Relationships & Foreign Key References:

- Referenced by `sales_orders.customer_id` (One-to-Many)

Table 5: sales_orders

Description: Top-level transactional head records for commercial sales, monetary totals, tracking dates, and terms mapping.

Column Name	Data Type	Key Type	Description / Constraints
order_id	int4	Primary Key	Unique auto-generated sequential transaction document key.
customer_id	int4	Foreign Key	References customers.customer_id establishing procurement owner.
order_date	timestampz	-	The exact timestamp of original transaction confirmation.
status	varchar	-	State variable monitoring lifecycle stage (e.g., 'PENDING', 'SHIPPED', 'PAID').
subtotal	numeric	-	Aggregated monetary value of all nested lines before tax.
tax_amount	numeric	-	Total cumulative taxation applied directly across order parameters.
total_amount	numeric	-	Net financial payload due for processing (Subtotal + Taxes).
payment_terms	varchar	-	Invoicing configuration terms (e.g., 'NET30', 'COD').
sales_rep_id	int4	Foreign Key	References employees.employee_id defining internal sales lead.

Relationships & Foreign Key References:

- Many-to-One relationship to **customers.customer_id**
- Many-to-One relationship to **employees.employee_id**
- Referenced by **order_items.order_id** (One-to-Many, cascade context)

Table 6: order_items

Description: Fine-grained transactional sub-lines itemizing individual items, quantities, discounts, and tallies linked to an order header.

Column Name	Data Type	Key Type	Description / Constraints
<code>item_id</code>	int4	Primary Key	Unique localized identifier for the line row entry.
<code>order_id</code>	int4	Foreign Key	References parent invoice document via <code>sales_orders.order_id</code> .
<code>product_id</code>	int4	Foreign Key	References purchased stock detail via <code>products.product_id</code> .
<code>quantity</code>	int4	-	The exact count of items demanded in this line segment.
<code>unit_price</code>	numeric	-	Snapshot value of the per-unit price applied for this sale.
<code>discount_percent</code>	numeric	-	Percentage deduction multiplier negotiated for the line item.
<code>line_total</code>	numeric	-	Calculated bottom-line value after quantitative multiplication and deductions.

Relationships & Foreign Key References:

- Many-to-One relationship to `sales_orders.order_id`
- Many-to-One relationship to `products.product_id`

Table 7: employees

Description: Internal payroll, department mapping, role tracking, and identity ledger for human resource management.

Column Name	Data Type	Key Type	Description / Constraints
<code>employee_id</code>	int4	Primary Key	Unique sequential indicator representing an internal workforce staff profile.
<code>first_name</code>	varchar	-	Legal given name of the worker.
<code>last_name</code>	varchar	-	Legal family surname of the worker.
<code>email</code>	varchar	Unique	Corporate email identifier matching organization records.
<code>hire_date</code>	date	-	Official commencement date of employment.
<code>dept_id</code>	int4	Foreign Key	References <code>departments.dept_id</code> for functional hierarchy assignment.
<code>job_title</code>	varchar	-	Work title or standard operational designation.
<code>salary</code>	numeric	-	Base compensation amount structured for payroll processing.
<code>is_active</code>	bool	-	Workforce availability state flag.

Relationships & Foreign Key References:

- Many-to-One relationship to `departments.dept_id`
- Referenced by `departments.manager_id` (One-to-Many / Self-Referencing structure)
- Referenced by `sales_orders.sales_rep_id` (One-to-Many)
- Referenced by `payslips.employee_id` (One-to-Many)

Table 8: departments

Description: Organizational business unit structural divisions linked directly to specific leadership nodes.

Column Name	Data Type	Key Type	Description / Constraints
dept_id	int4	Primary Key	Unique ID code for the business division.
dept_name	varchar	Unique	Descriptive name of the corporate vertical group.
manager_id	int4	Foreign Key	References employees.employee_id to assign a direct department head.

Relationships & Foreign Key References:

- Many-to-One relationship to **employees.employee_id** (Manager mapping)
- Referenced by **employees.dept_id** (One-to-Many operational link)

Table 9: accounts_chart (Chart of Accounts)

Description: Definitional asset, liability, equity, and operational ledger balance structure definitions.

Column Name	Data Type	Key Type	Description / Constraints
account_code	varchar	Primary Key	Unique bookkeeping ledger numeric/alphanumeric indexing code string.
account_name	varchar	-	Common corporate definition title for balancing.
account_type	varchar	-	Category marker (e.g., 'ASSET', 'LIABILITY', 'REVENUE', 'EXPENSE').
currency	varchar	-	Standard base ISO currency label (e.g., 'USD', 'EUR').

Relationships & Foreign Key References:

- Referenced by **general_ledger.account_code** (One-to-Many core booking link)

Table 10: general_ledger

Description: Immutable core accounting transactions log capturing double-entry financial impacts.

Column Name	Data Type	Key Type	Description / Constraints
<code>entry_id</code>	uuid	Primary Key	Globally unique ID string generated per ledger mutation row.
<code>account_code</code>	varchar	Foreign Key	References <code>accounts_chart.account_code</code> .
<code>entry_date</code>	timestampz	-	The authorized calendar date of formal accounting recognition.
<code>debit</code>	numeric	-	Monetary debit adjustment amount parameters.
<code>credit</code>	numeric	-	Monetary credit adjustment amount parameters.
<code>reference_order_id</code>	int4	Foreign Key	Optional connection reference payload directly tracking a <code>sales_orders.order_id</code> .
<code>description</code>	text	-	Contextual notes describing the business event requiring balancing.

Relationships & Foreign Key References:

- Many-to-One relationship to `accounts_chart.account_code`
- Many-to-One relationship to `sales_orders.order_id` via `reference_order_id`

Table 11: Auxiliary Infrastructure (payslips & audit_logs)

Description: Supporting system tables that handle automated recurring human resources payouts and metadata system activity monitoring metrics.

- **payslips table:** Contains `payslip_id` (int4, PK), `employee_id` (int4, FK pointing to `employees.employee_id`), pay period bounds, gross amounts, net calculations, and disbursement dates.
- **audit_logs table:** System event log recording security changes, tracking tables mutated, active operator contexts, timestamps, and row transaction payloads.